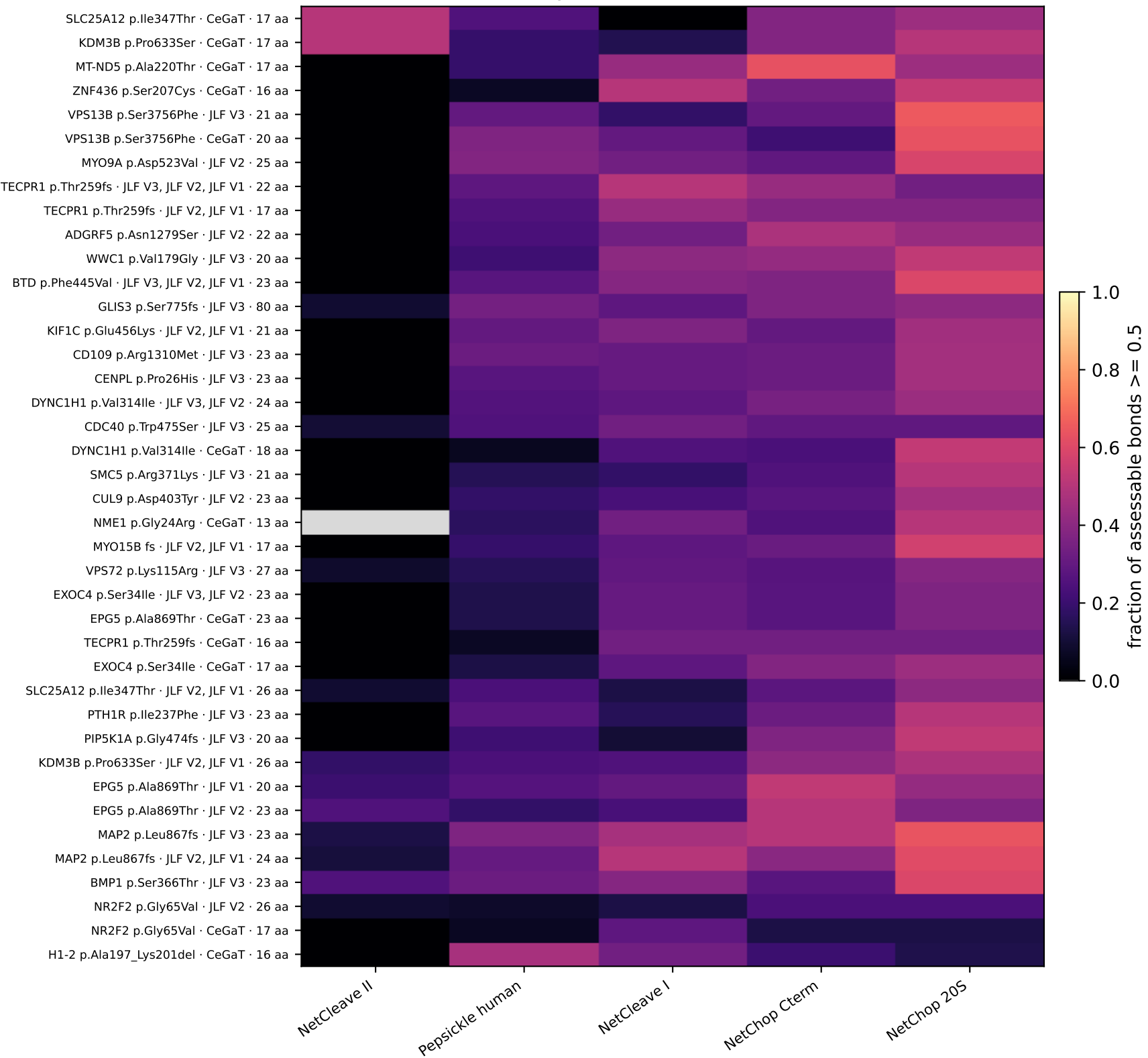
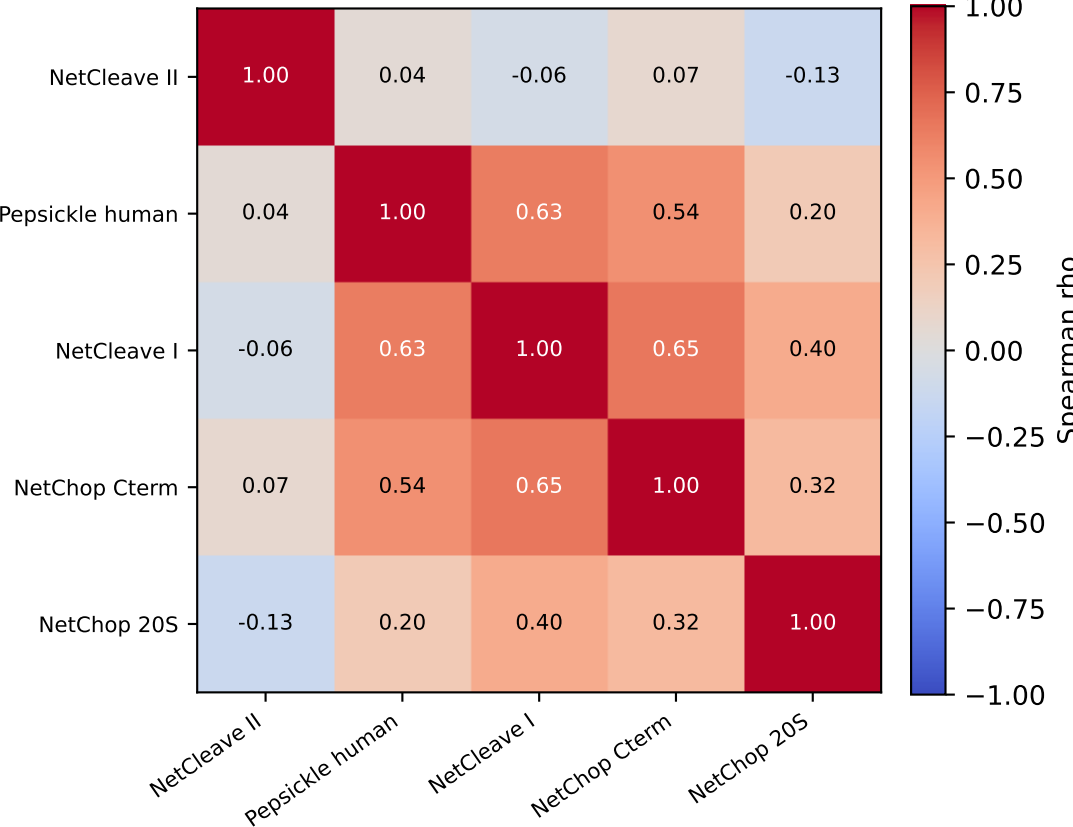


Agreement map for the vaccine SLP cleavage atlas

SLPs clustered by within-model candidate-site fraction



Predictor agreement at shared peptide bonds



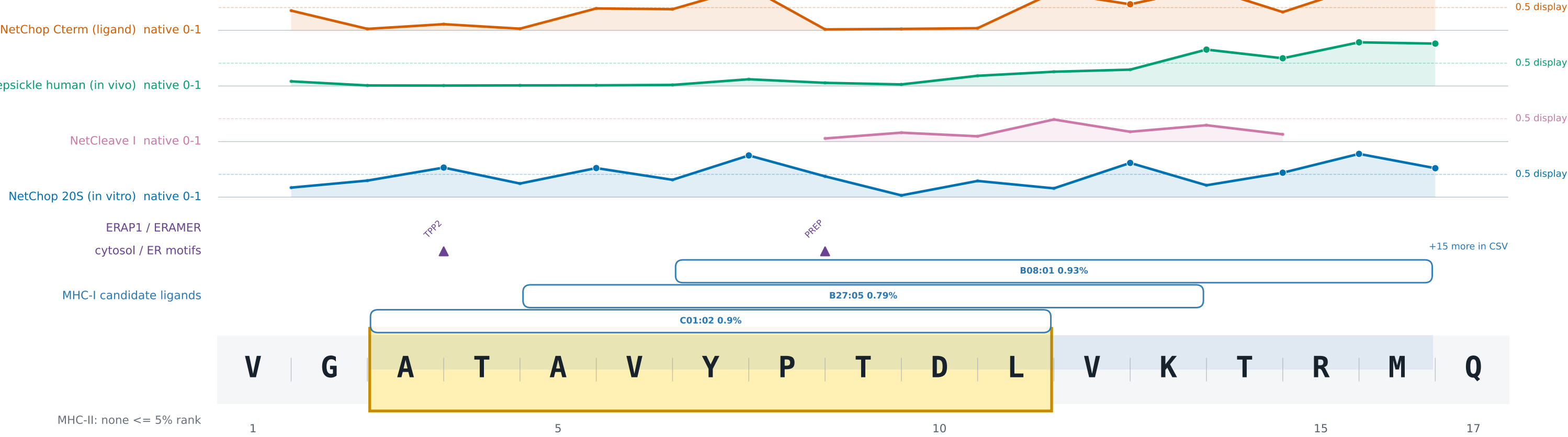
Clustering organizes the atlas; it does not create an ensemble probability. Sequence pages follow the left-panel SLP order. The figure shows human-only Pepsickle, both biologically distinct NetChop modes, NetCleave-I, and NetCleave-II; the near-redundant all-mammal Pepsickle track remains in the tables. The left panel uses a common display threshold on native 0-1 outputs. The right panel correlates native scores only where both models assess the same bond.

SLC25A12 p.Ile347Thr | CeGaT | 17 aa

disclosed minimal epitope ATAVYPTDL at 3-11

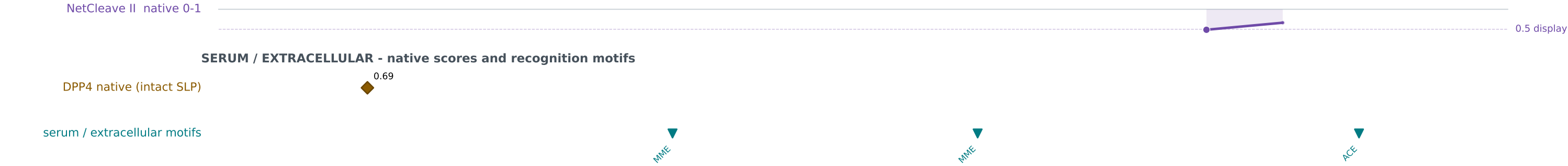
INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

KDM3B p.Pro633Ser | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

XPNPEP1
DPP8+DPP9+PREP
TPP2

PREP

+9 more in CSV

MHC-I candidate ligands

C01:02 0.22%

C07:01 0.86%

B27:05 1.2%

P | P | K | L | S | R | E | E | S | S | N | P | F | L | A | F | V

MHC-II: none <= 5% rank

1 5 10 15 17

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

serum / extracellular motifs

XPNPEP2 FAP MME MME MME ACE+MME

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

MT-ND5 p.Ala220Thr | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER

cytosol / ER motifs

TPP2

+2 more in CSV

MHC-I candidate ligands

C01:02 0.72%

C01:02 0.34%

C01:02 1.3%

P | L | L | G | L | L | L | A | T | A | G | K | S | A | Q | L | G

MHC-II: none <= 5% rank

1

5

10

15

17

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.026

serum / extracellular motifs

MME

MME

MME

MME

MME

ACE+MME

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

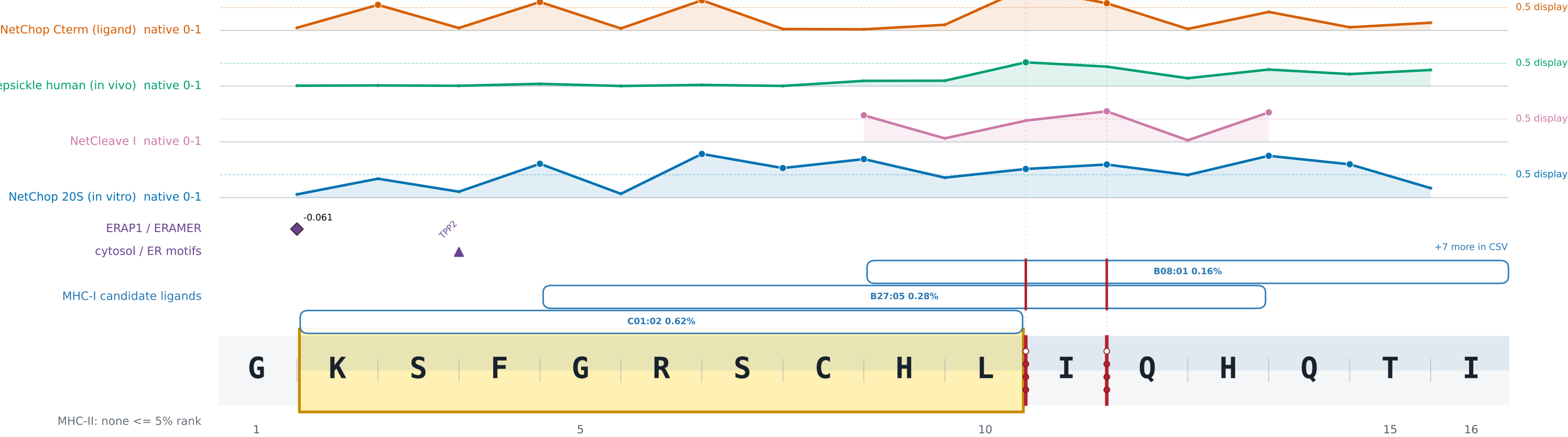
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

ZNF436 p.Ser207Cys | CeGaT | 16 aa

disclosed minimal epitope KSFGRSCHL at 2-10

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

serum / extracellular motifs

MME

MME

MME

ACE

MME

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

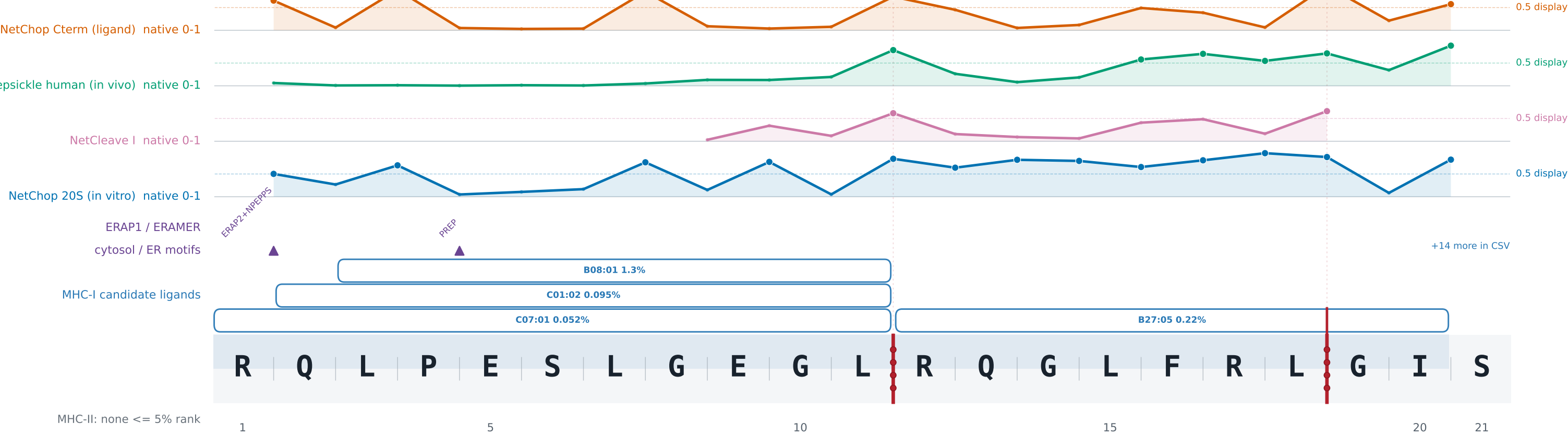
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

VPS13B p.Ser3756Phe | JLF V3 | 21 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

VPS13B p.Ser3756Phe | CeGaT | 20 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

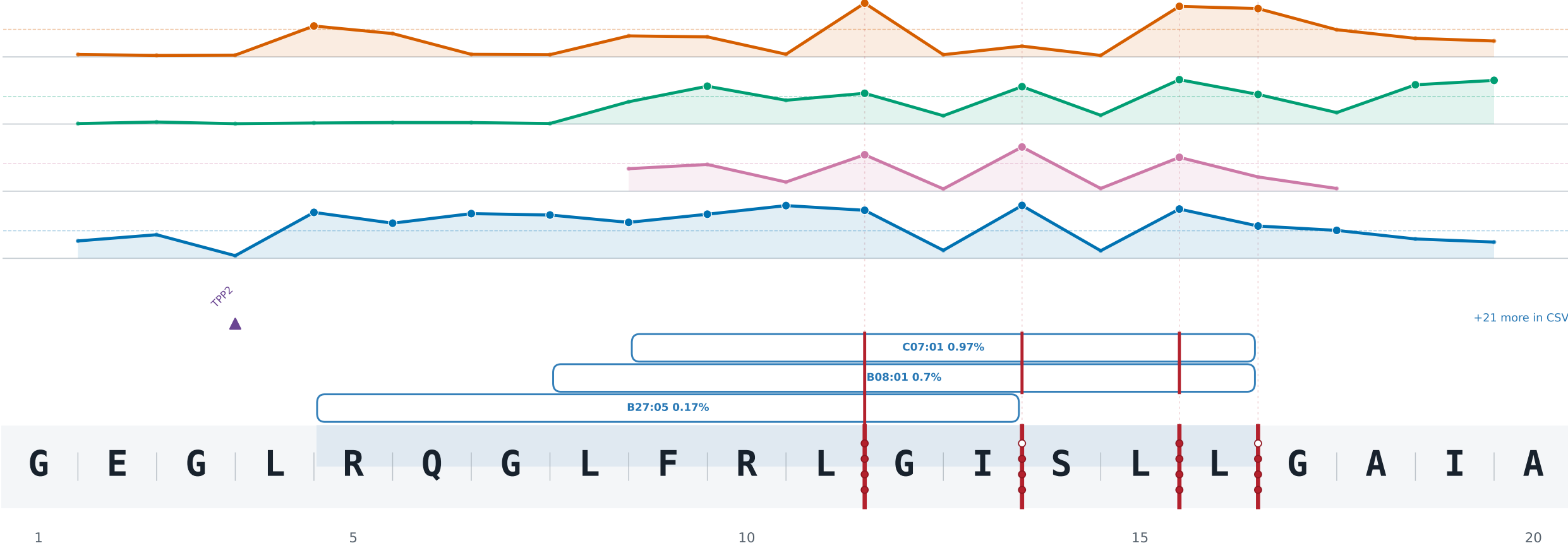
ERAP1 / ERAMER
cytosol / ER motifs

TPP2

+21 more in CSV

MHC-I candidate ligands

MHC-II: none <= 5% rank



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.02

serum / extracellular motifs

MME

MME

MME

MME

MME

MME

MME

ACE+MME

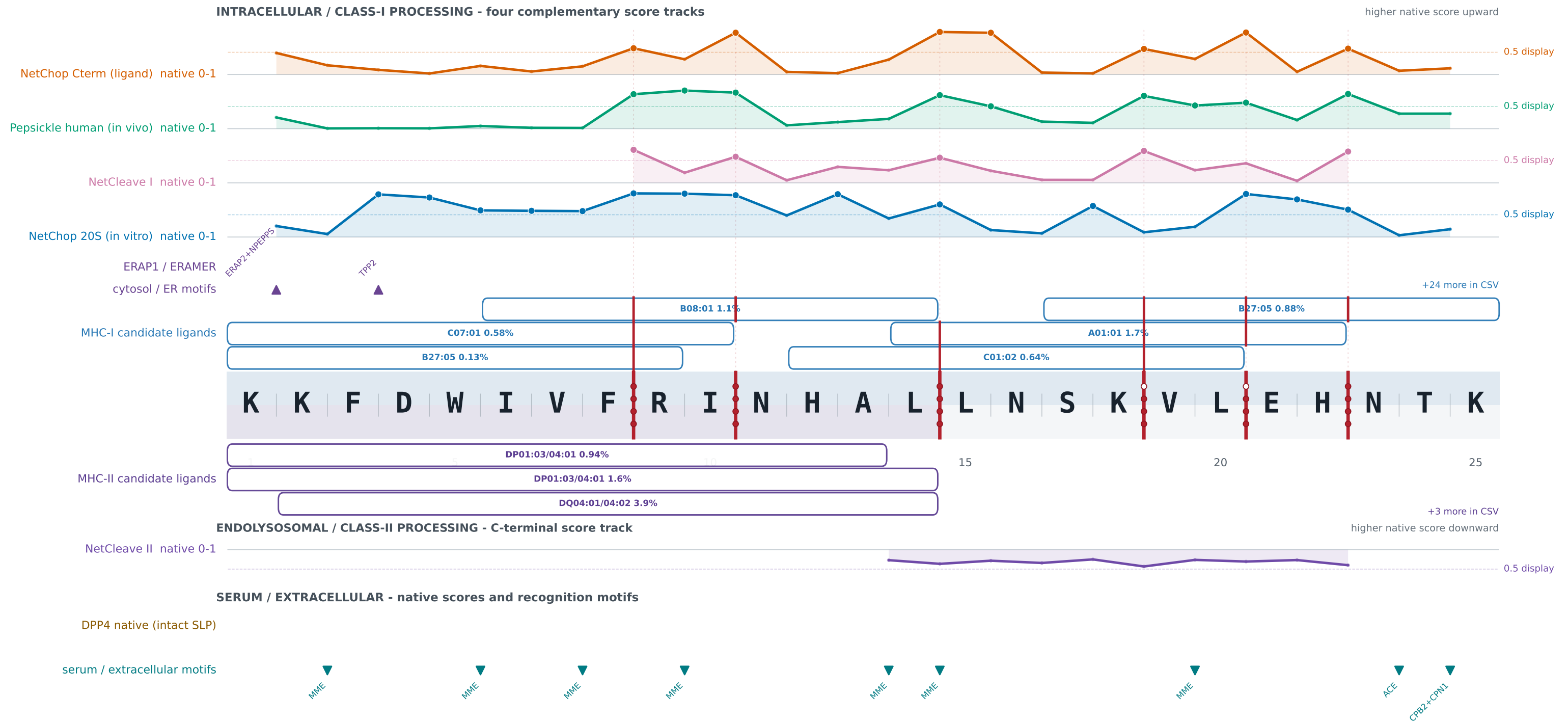
SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

MYO9A p.Asp523Val | JLF V2 | 25 aa

minimal epitope/window not disclosed for this SLP



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

TECPR1 p.Thr259fs | JLF V3, JLF V2, JLF V1 | 22 aa

disclosed minimal epitope TIPAAPVGL at 9-17

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

TPP2

PREP

PREP

+12 more in CSV

MHC-I candidate ligands

C07:01 1.9%

C01:02 0.095%

B27:05 0.48%

G | S | E | E | A | S | T | L | T | I | P | A | A | P | V | G | L | R | W | L | V | R

MHC-II: none <= 5% rank

1 5 10 15 20 22

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

-0.085

serum / extracellular motifs

MME

MME

MME

MME

ACE

CPB2+CPN1

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

TECPR1 p.Thr259fs | JLF V2, JLF V1 | 17 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

NPEPPS

TPP2

PREP

+2 more in CSV

MHC-I candidate ligands

C07:01 0.25%

B27:05 1%

B27:05 0.29%

T S R W E S A W S G L S P R T G R

MHC-II candidate ligands

DQ04:01/04:02 1.9%

DQ04:01/04:02 1.4%

DQ04:01/04:02 1.3%

+2 more in CSV

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

serum / extracellular motifs

MME

ACE

CPB2+CPN1

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

ADGRF5 p.Asn1279Ser | JLF V2 | 22 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

NPEPP5

TPP2

+4 more in CSV

MHC-I candidate ligands

B08:01 0.36%

B08:01 0.35%

B27:05 1.4%

B27:05 1.9%

B27:05 0.38%

V | Q | E | A | L | L | S | K | F | S | L | S | R | W | S | S | Q | H | S | K | S | T

MHC-II candidate ligands

DQ04:01/04:02 1.1%

DQ04:01/04:02 0.79%

DQ04:01/04:02 0.98%

+11 more in CSV

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.019

serum / extracellular motifs

MME

MME

MME

MME

ACE

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

WWC1 p.Val179Gly | JLF V3 | 20 aa

disclosed minimal epitope GHQLQHELQF at 8-16

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

NPEPP5

TPP2

+10 more in CSV

MHC-I candidate ligands

C07:01 0.082%

C01:02 0.67%

B27:05 0.56%

N | K | L | K | R | E | M | G | H | L | Q | H | E | L | Q | F | K | E | R | G

MHC-II: none <= 5% rank

1

5

10

15

20

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.065

serum / extracellular motifs

MME

MME

MME

MME

ACE

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

BTD p.Phe445Val | JLF V3, JLF V2, JLF V1 | 23 aa

disclosed minimal epitope GVVDGLHTV at 12-20

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

XPNPEP1+ERAP2
DPP8+DPP9+PHEP

TPP2

+14 more in CSV

MHC-I candidate ligands

B08:01 0.83%

A01:01 0.7%

C07:01 1.7%

C01:02 0.052%

B08:01 1.1%

B27:05 1.8%

R | P | T | L | S | K | E | L | Y | A | L | G | V | V | D | G | L | H | T | V | H | G | T

MHC-II candidate ligands

DP01:03/04:01 4.7%

DP01:03/04:01 5%

DP01:03/04:01 3.9%

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

3.1

serum / extracellular motifs

XPNPEP2

FAP

MME

MME

MME

MME

MME

ACE

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

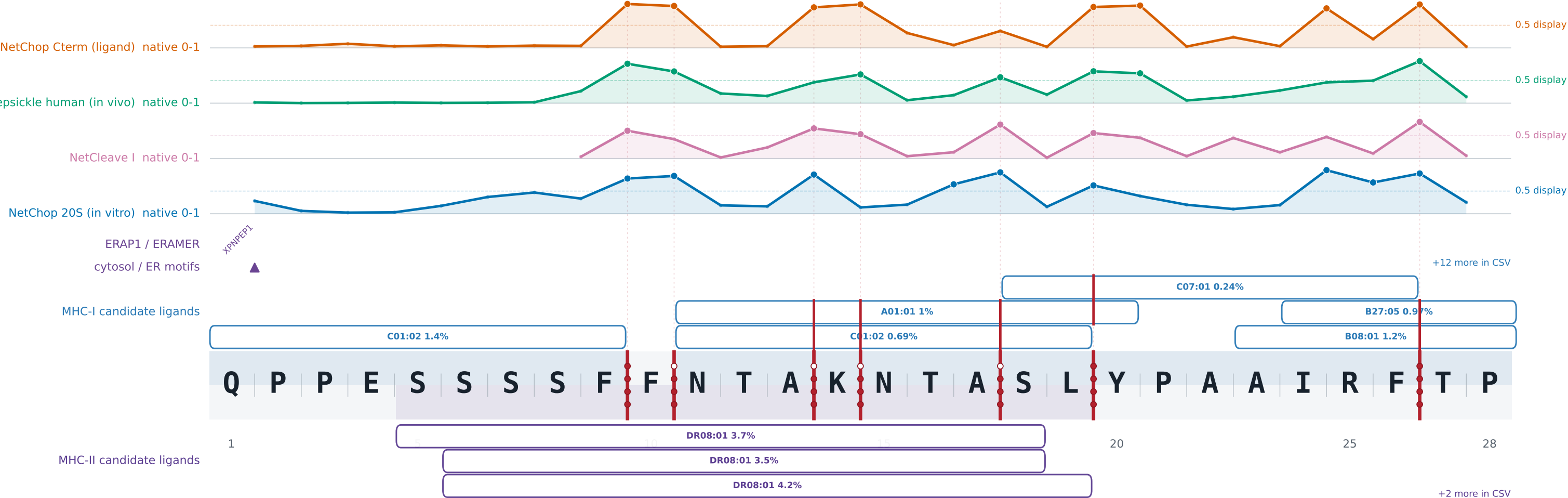
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 1/3, bonds 1-27

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

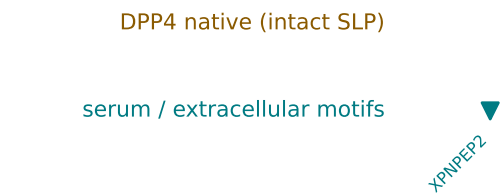


ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

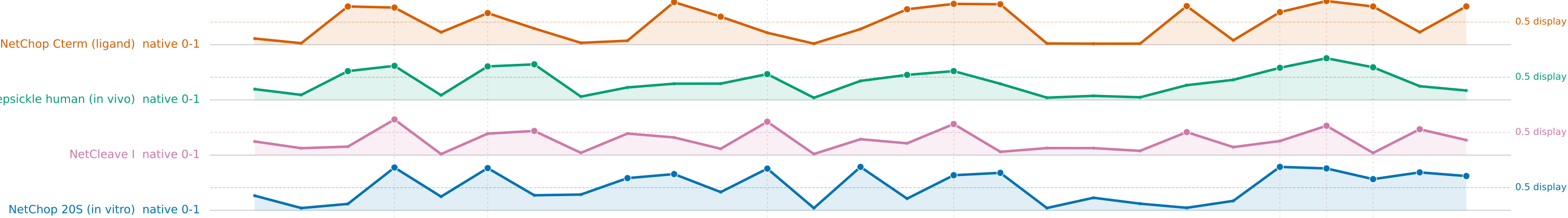
MHC WINDOWS $\leq 2\%$ / $\leq 5\%$ rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 2/3, bonds 28-54

minimal epitope/window not disclosed for this SLP

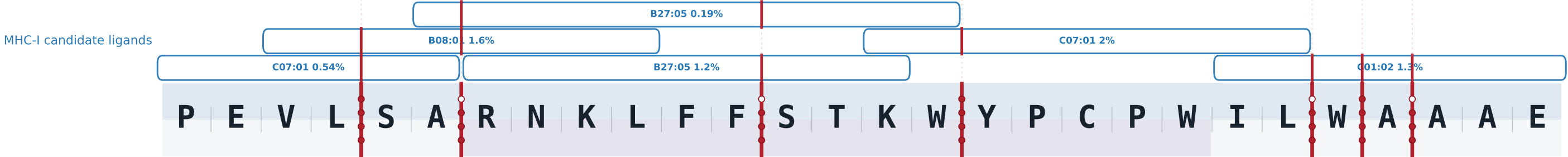
INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

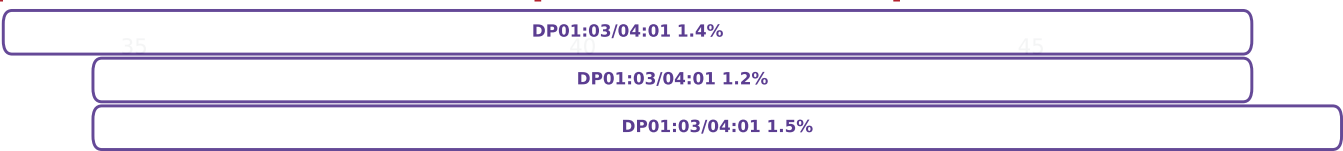


ERAP1 / ERAMER
cytosol / ER motifs

+8 more in CSV

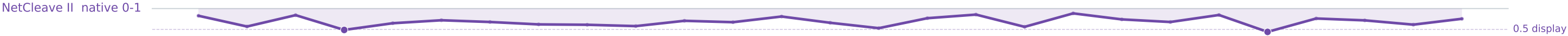


MHC-II candidate ligands



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

+16 more in CSV
higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

serum / extracellular motifs

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

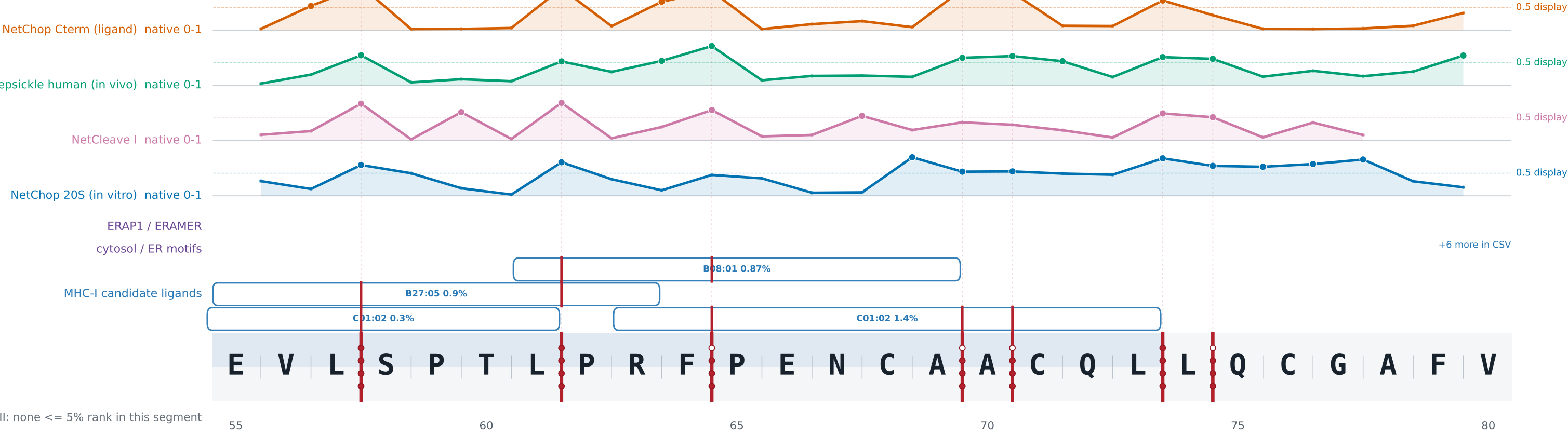
MHC WINDOWS $\leq 2\%/\leq 5\%$ rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 3/3, bonds 55-79

minimal epitope/window not disclosed for this SLP

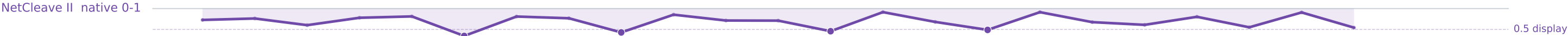
INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

serum / extracellular motifs

ACE

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

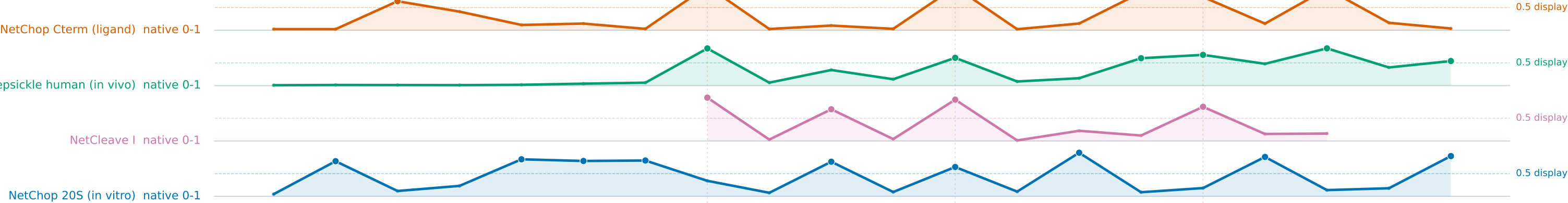
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

KIF1C p.Glu456Lys | JLF V2, JLF V1 | 21 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

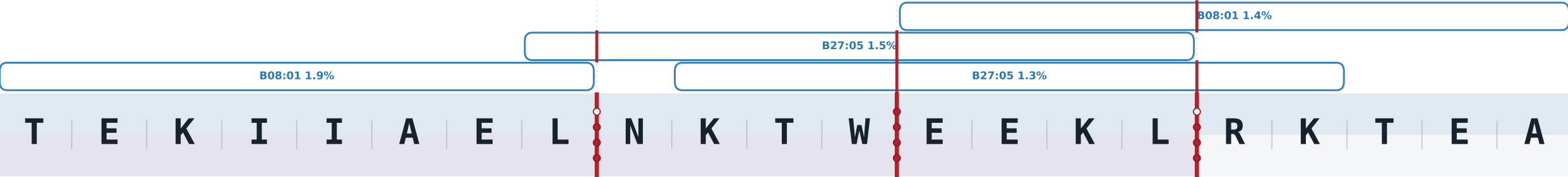


NPEPPS

TPP2

ERAP1 / ERAMER
cytosol / ER motifs

MHC-I candidate ligands



MHC-II candidate ligands



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.0002

serum / extracellular motifs

MME

MME

MME

MME

ACE

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

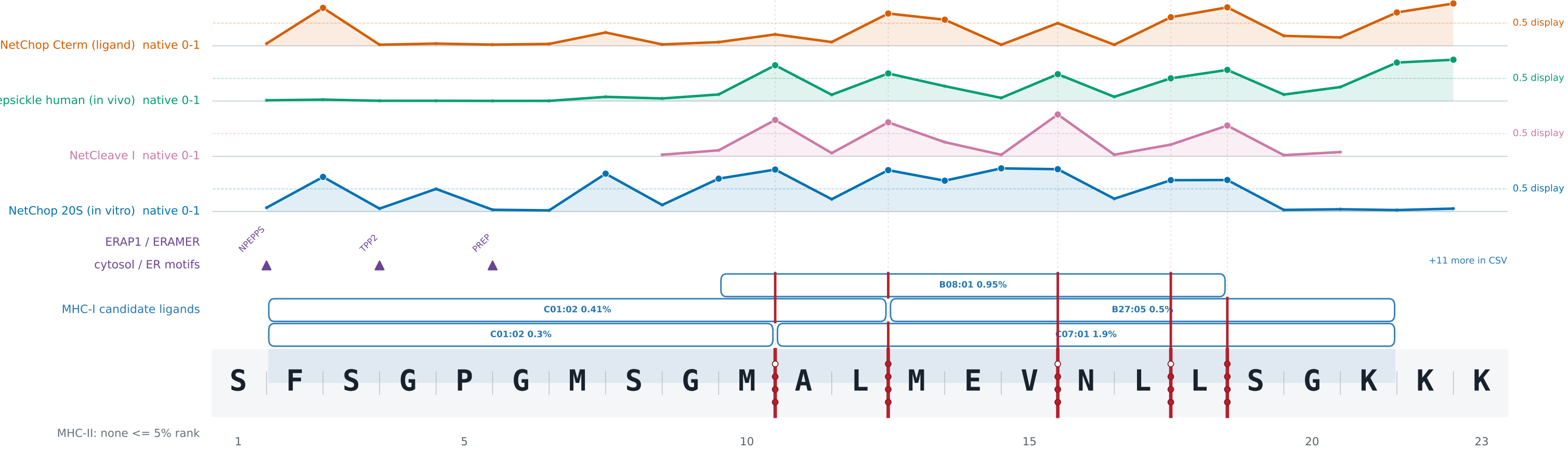
MHC WINDOWS $\leq 2\% / \leq 5\%$ rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

CD109 p.Arg1310Met | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP

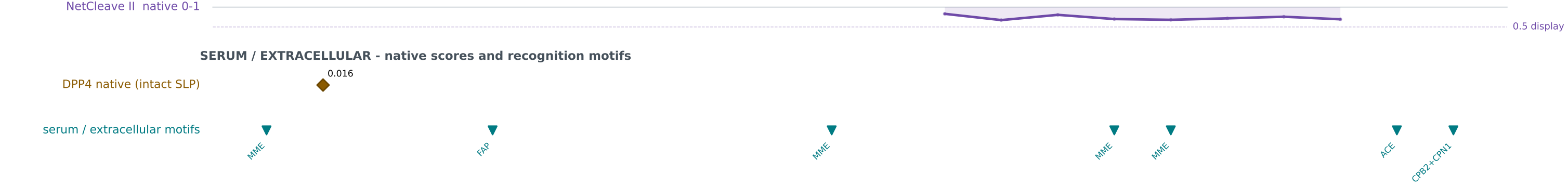
INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

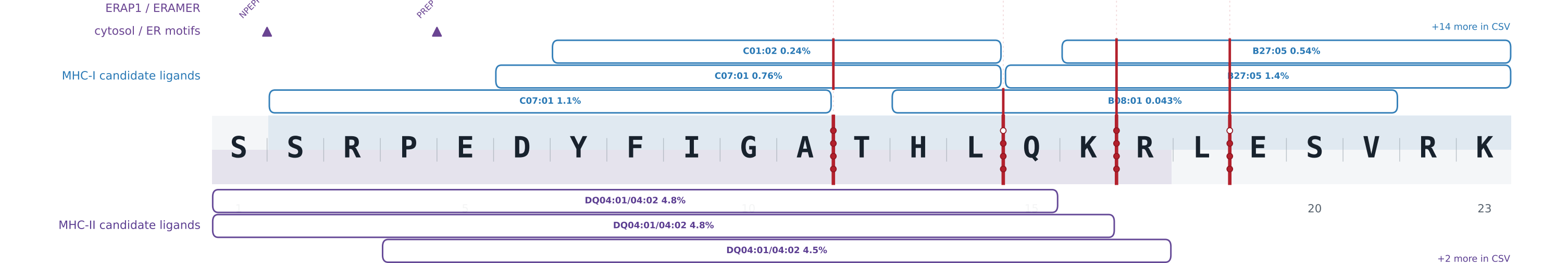
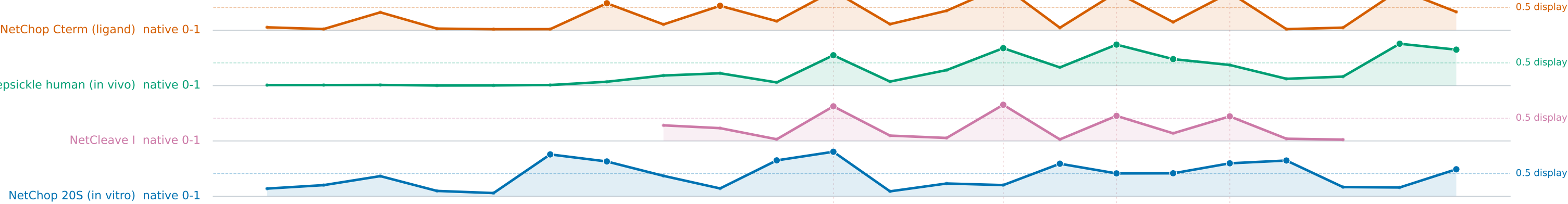
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

CENPL p.Pro26His | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

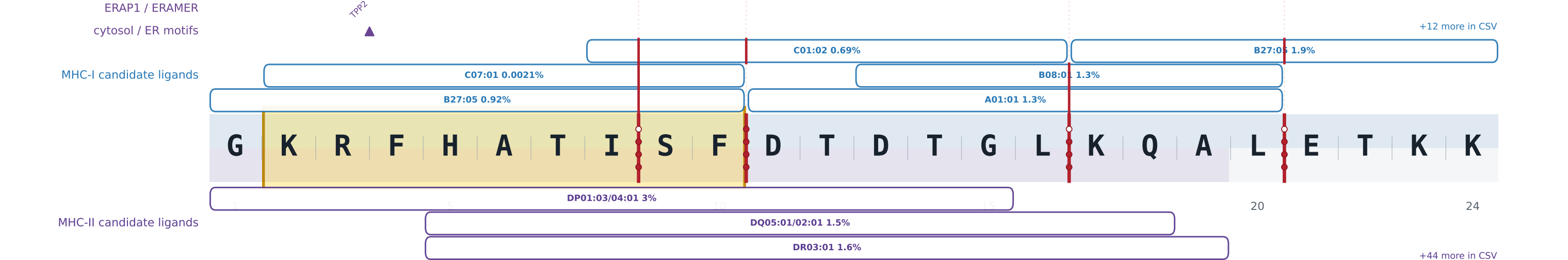
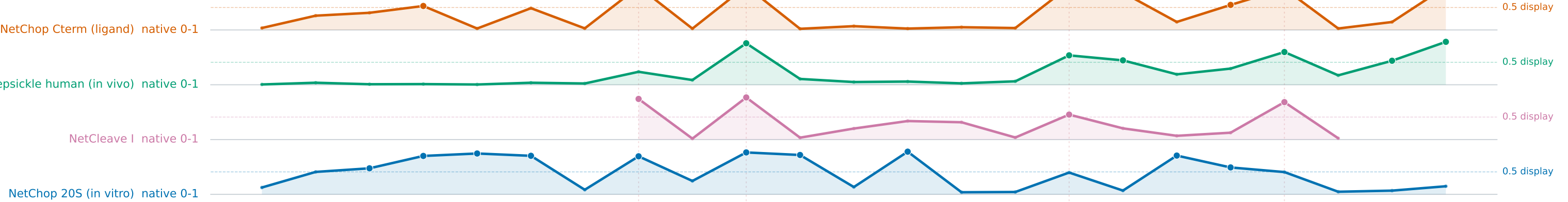
MHC WINDOWS $\leq 2\% / \leq 5\%$ rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

DYNC1H1 p.Val314Ile | JLF V3, JLF V2 | 24 aa

disclosed minimal epitope KRFHATISF at 2-10

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

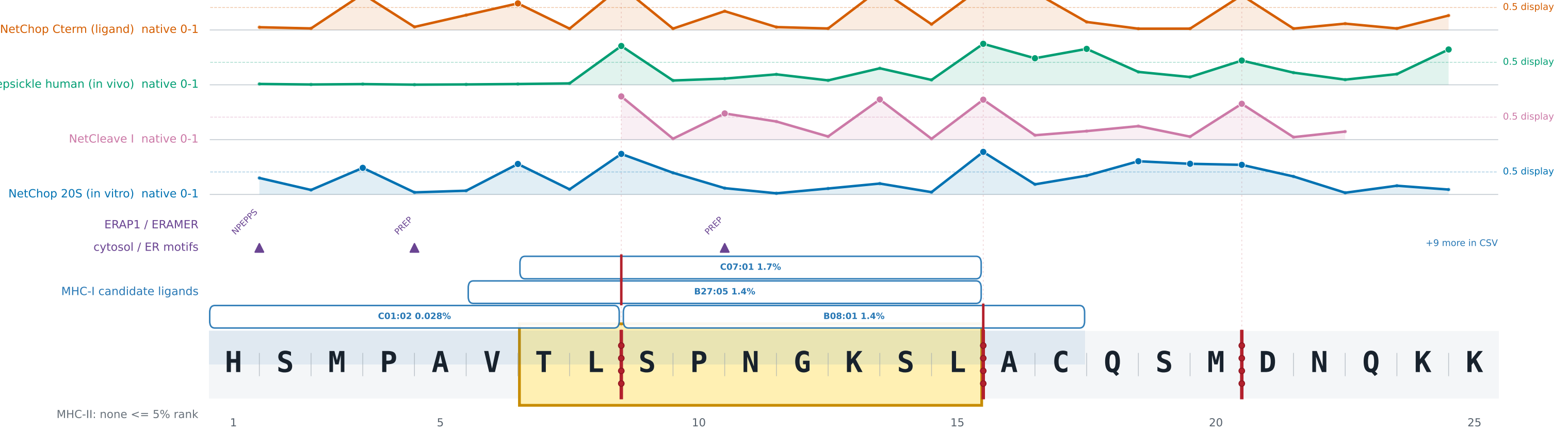
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

CDC40 p.Trp475Ser | JLF V3 | 25 aa

disclosed minimal epitope TLSPNGKSL at 7-15

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

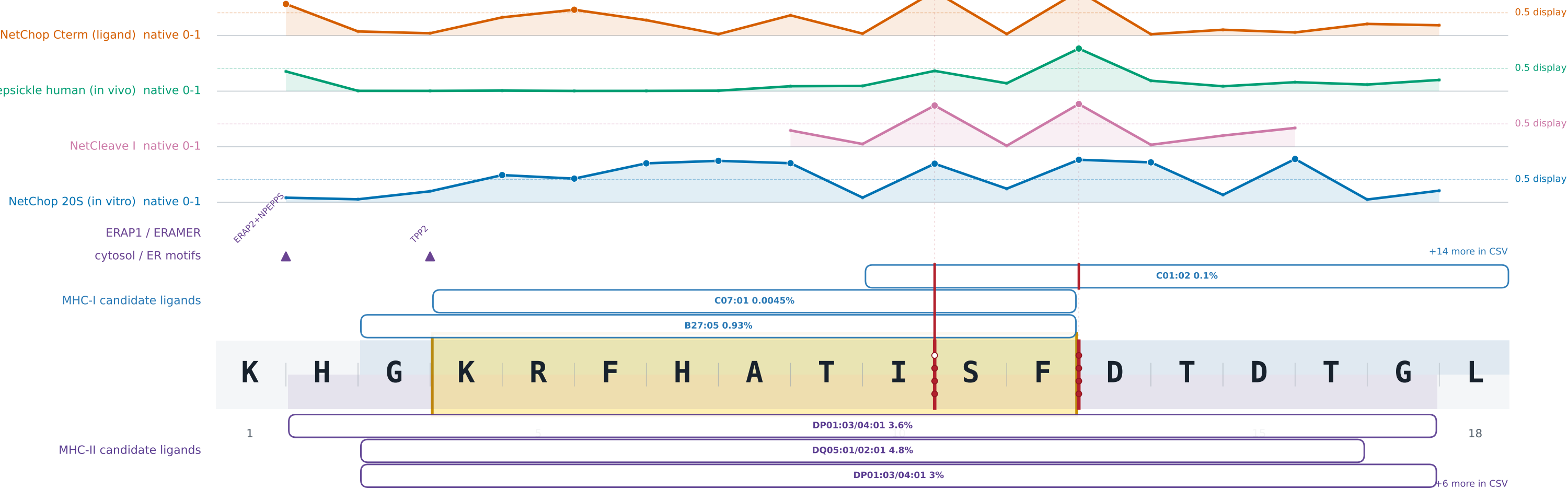
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

DYNC1H1 p.Val314Ile | CeGaT | 18 aa

disclosed minimal epitope KRFHATISF at 4-12

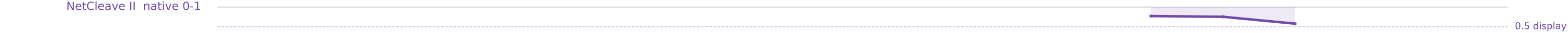
INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS $\leq 2\%$ / $\leq 5\%$ rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

SMC5 p.Arg371Lys | JLF V3 | 21 aa

disclosed minimal epitope RQKRIGNTR at 2-10

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

NPEPP5

TPP2

+29 more in CSV

MHC-I candidate ligands

C07:01 0.31%

B27:05 0.58%

B27:05 0.54%

A01:01 0.87%

B27:05 0.2%

C01:02 0.019%

D R Q K R I G N T R K M I E D L Q N E L K

MHC-II candidate ligands

DR08:01 3.2%

15

20

21

DR08:01 2.9%

DR03:01 3.4%

+10 more in CSV

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

serum / extracellular motifs

ENPEP

MME

MME

MME

ACE+MME

CPB2+CPH1

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

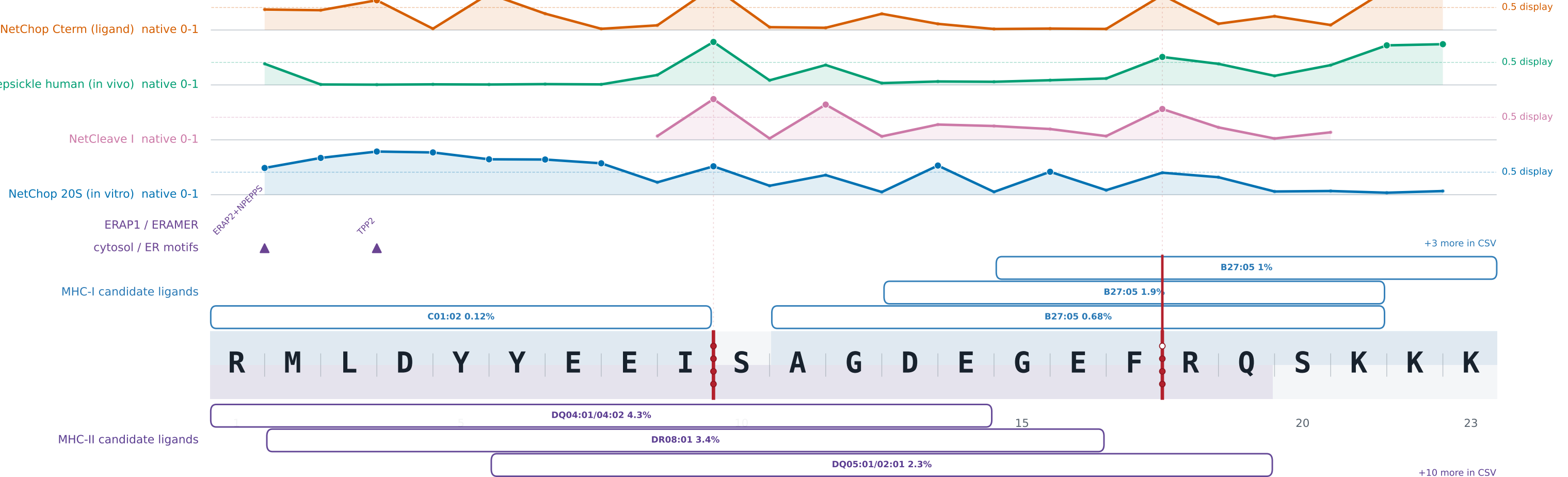
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

CUL9 p.Asp403Tyr | JLF V2 | 23 aa

minimal epitope/window not disclosed for this SLP

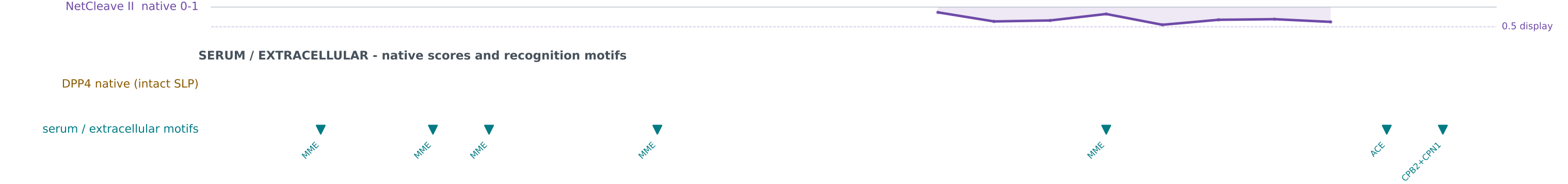
INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

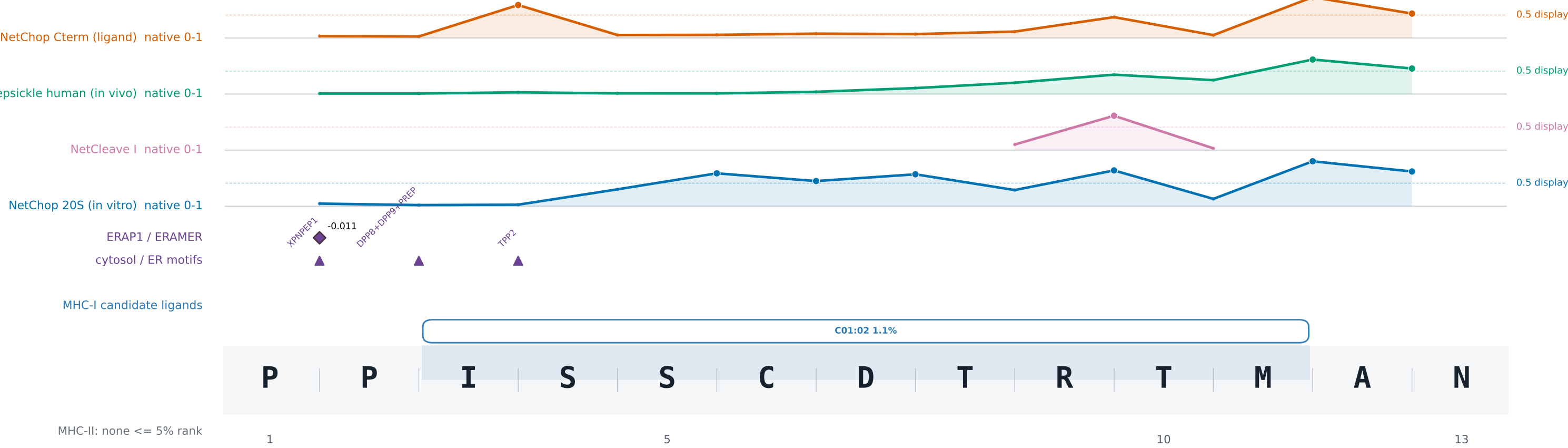
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

NME1 p.Gly24Arg | CeGaT | 13 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

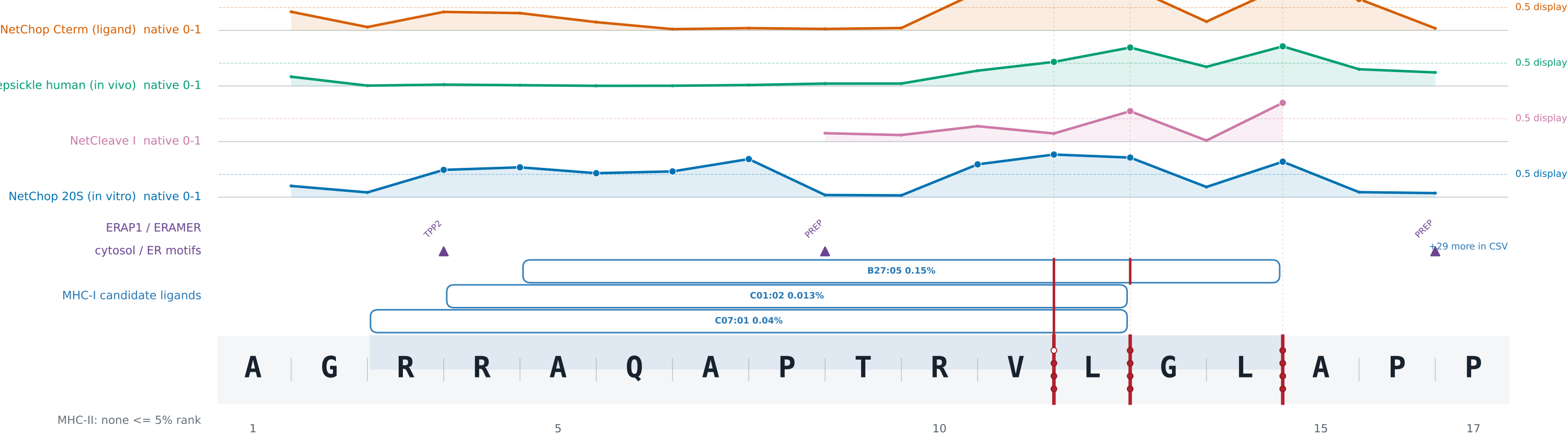
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

MYO15B fs | JLF V2, JLF V1 | 17 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

serum / extracellular motifs

ANPEP

MME

MME

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

VPS72 p.Lys115Arg | JLF V3 | 27 aa

disclosed minimal epitope ERALLPLEL at 9-17

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

TPP2

PREP

+9 more in CSV

MHC-I candidate ligands

C01:02 0.31%

C07:01 0.25%

B27:05 1.1%

G | S | S | Q | K | A | R | E | E | R | A | L | L | P | L | E | L | Q | D | D | G | S | D | S | R | K | S

1

5

10

15

DR03:01 2.7%

20

25

30

DR03:01 2.2%

DR03:01 2.3%

+7 more in CSV

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.34

serum / extracellular motifs

MME

MME

MME

MME

ACE

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

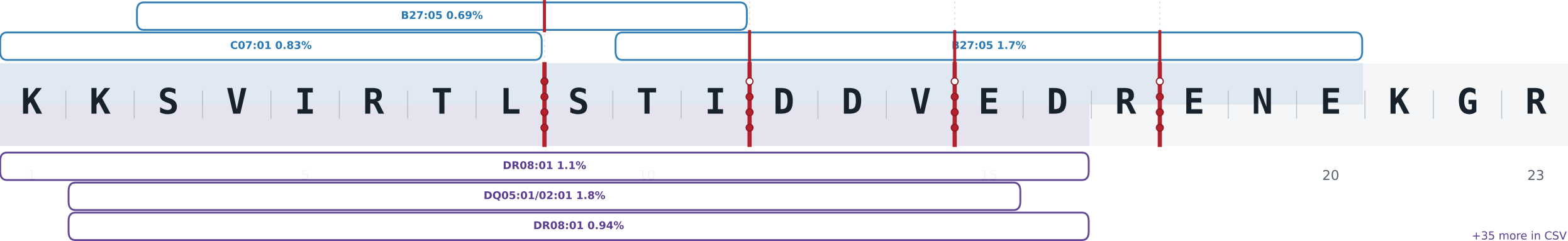
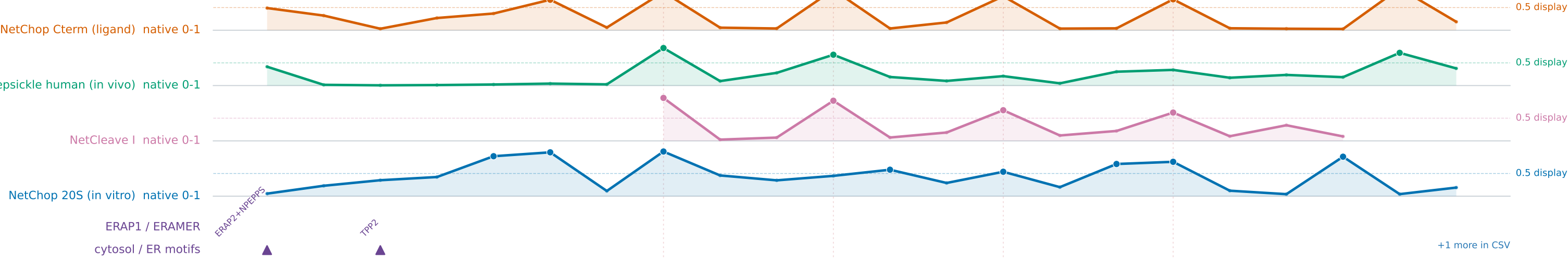
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

EXOC4 p.Ser34Ile | JLF V3, JLF V2 | 23 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

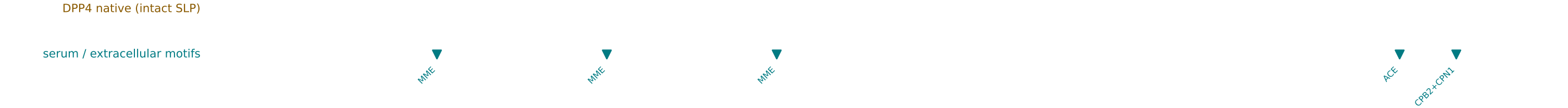


ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

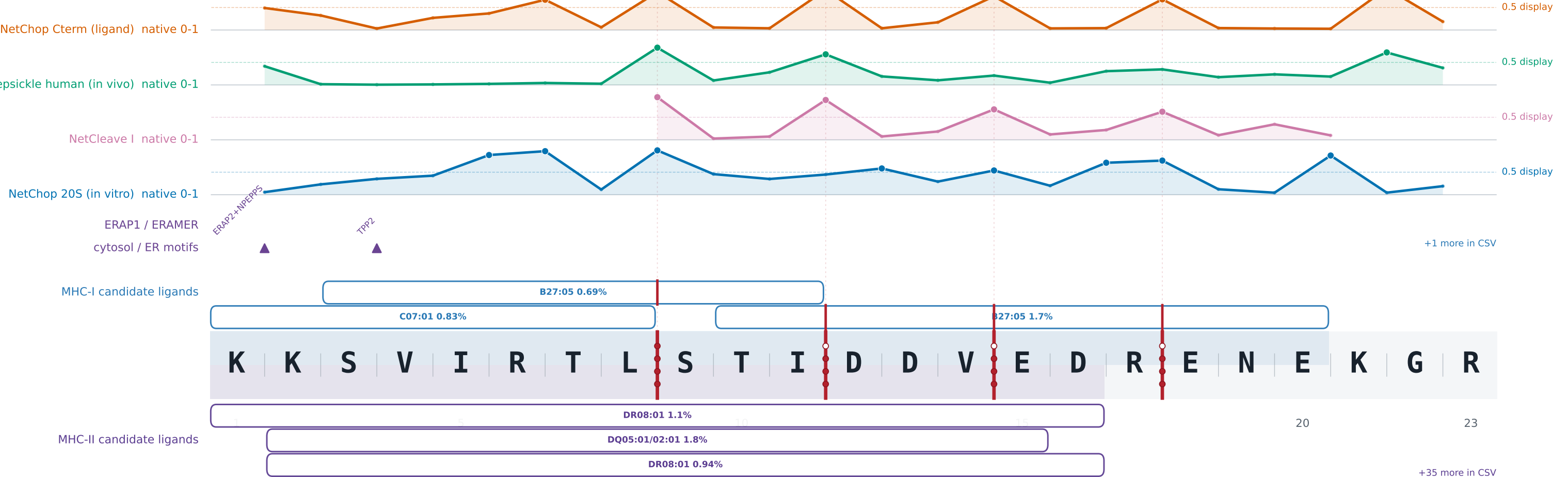
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

EPG5 p.Ala869Thr | CeGaT | 23 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS $\leq 2\%/\leq 5\%$ rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

TECPR1 p.Thr259fs | CeGaT | 16 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

NPEPPS
-0.18

TPP2

PREP

MHC-I candidate ligands

B27:05 0.8%

L

W

A

H

S

G

R

D

R

P

W

S

G

K

E

S

MHC-II: none <= 5% rank

1

5

10

15

16

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

-0.0099

serum / extracellular motifs

ACE

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

EXOC4 p.Ser34Ile | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

NPEPPS

TPP2

+2 more in CSV

MHC-I candidate ligands

B27:05 0.42%

B27:05 1.9%

S V I R T L S T I D D V E D R E N

MHC-II candidate ligands

DR08:01 2.8%

DQ05:01/02:01 2.2%

DQ05:01/02:01 2.1%

+12 more in CSV

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

-0.015

serum / extracellular motifs

MME

MME

MME

ACE

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

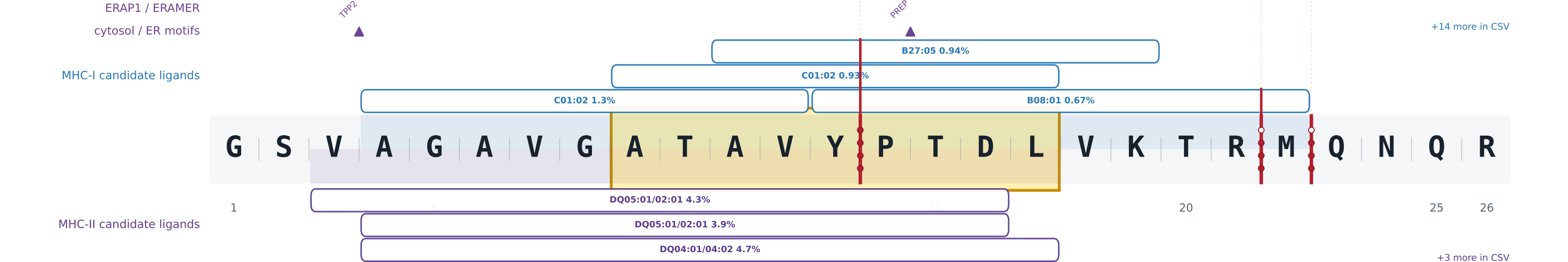
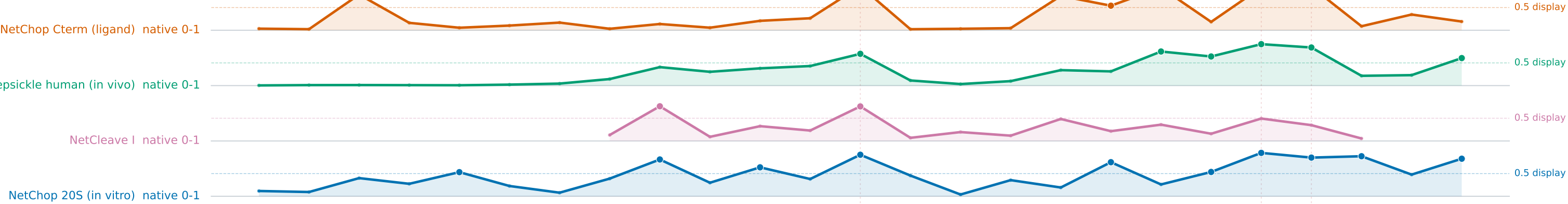
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

SLC25A12 p.Ile347Thr | JLF V2, JLF V1 | 26 aa

disclosed minimal epitope ATAVYPTDL at 9-17

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

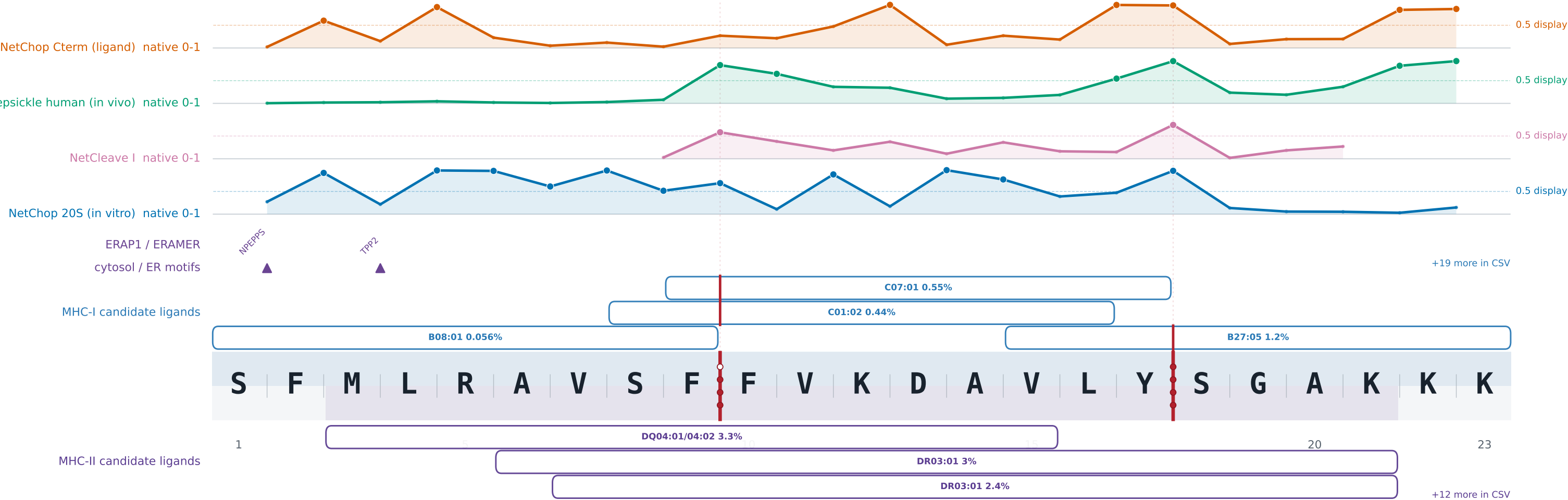
MHC WINDOWS $\leq 2\% / \leq 5\%$ rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

PTH1R p.Ile237Phe | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS $\leq 2\%/\leq 5\%$ rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

PIP5K1A p.Gly474fs | JLF V3 | 20 aa

disclosed minimal epitope AAPVATPAL at 7-15

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

NPEPPS

TPP2

PREP

PREP

+23 more in CSV

MHC-I candidate ligands

B08:01 0.36%

C01:02 0.036%

C07:01 0.2%

S

S

F

S

R

R

A

A

P

V

A

T

P

A

L

L

T

S

H

R

MHC-II: none <= 5% rank

1

5

10

15

20

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.62

serum / extracellular motifs

MME

MME

MME

ACE

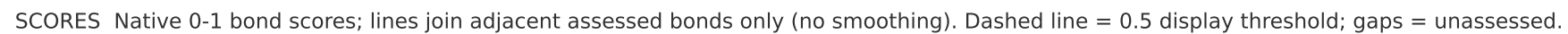
CPB2+CPN1

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

minimal epitope/window not disclosed for this SLP



RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

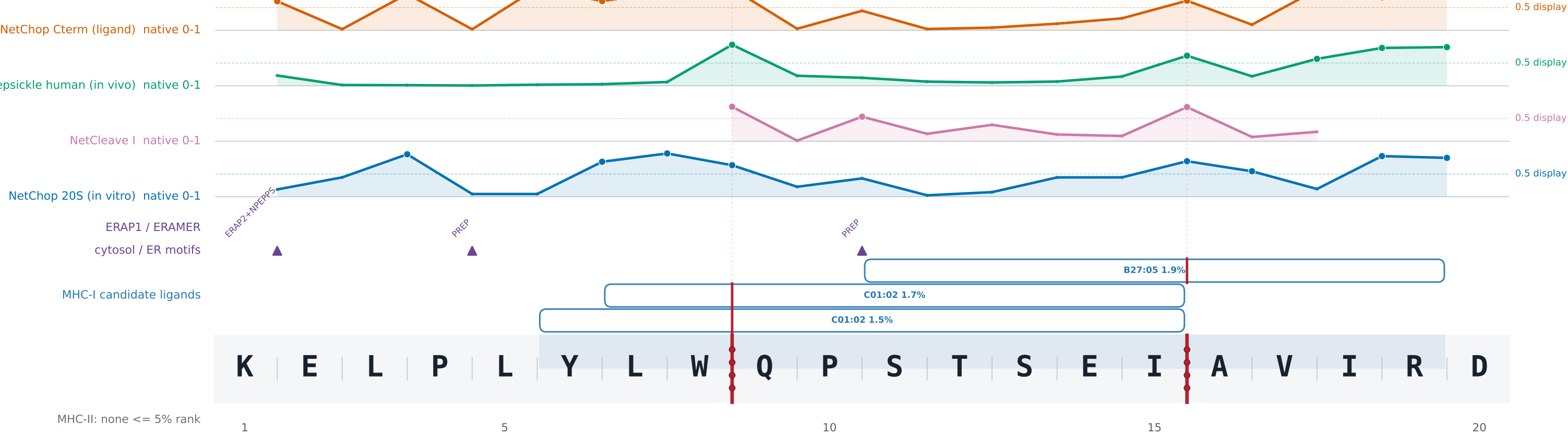
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

EPG5 p.Ala869Thr | JLF V1 | 20 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.011

serum / extracellular motifs

MME

MME

MME

MME

MME

MME

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

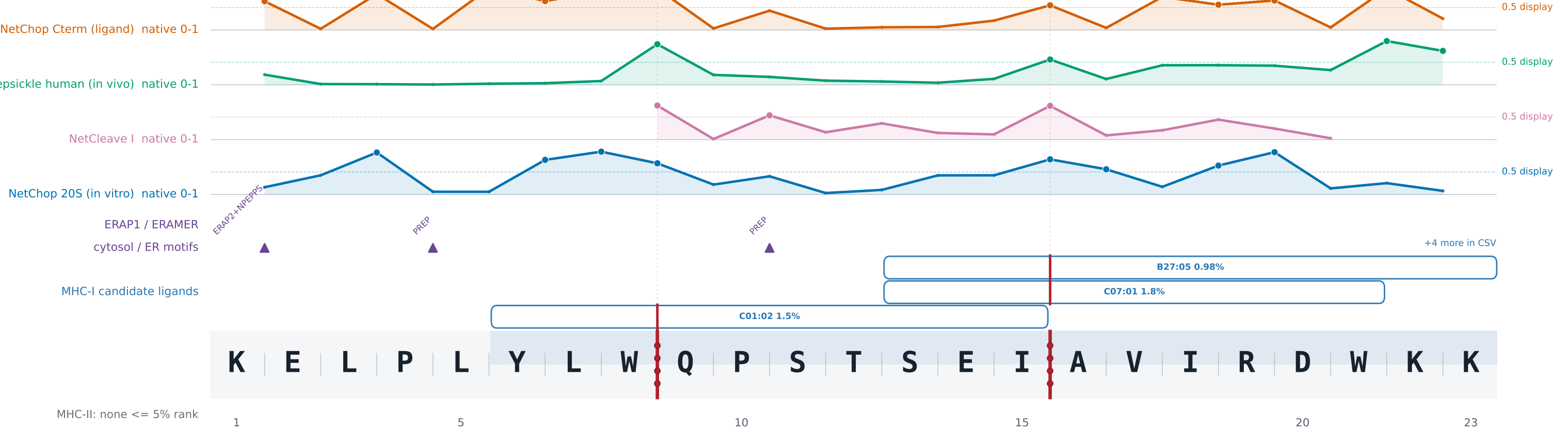
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

EPG5 p.Ala869Thr | JLF V2 | 23 aa

minimal epitope/window not disclosed for this SLP

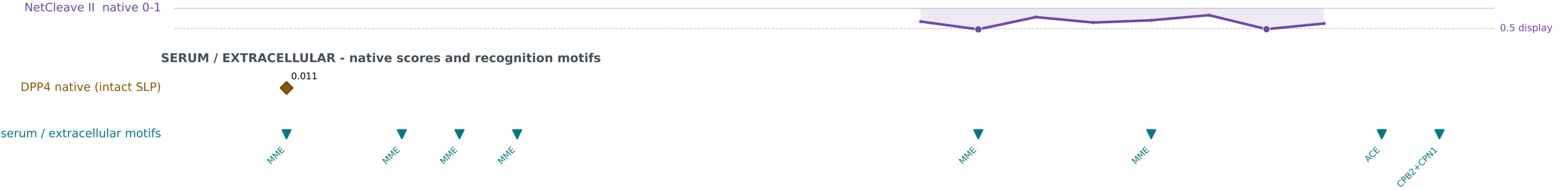
INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

MAP2 p.Leu867fs | JLF V3 | 23 aa

disclosed minimal epitope RVVPFTKAL at 9-17

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

ERAP2+MPEP5

TPP2

PREP

+31 more in CSV

MHC-I candidate ligands

C01:02 0.04%

C07:01 0.058%

B27:05 1.3%

B08:01 0.075%

K | T | V | R | I | Y | Q | G | R | V | V | P | F | T | K | A | L | M | I | K | F | E | E

MHC-II candidate ligands

DQ04:01/04:02 1.1%

DQ05:01/02:01 4.4%

DR08:01 3.5%

+8 more in CSV

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.49

serum / extracellular motifs

MME

MME

MME

MME

MME

MME

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

MAP2 p.Leu867fs | JLF V2, JLF V1 | 24 aa

disclosed minimal epitope RVVPFTKAL at 9-17

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

ERAP1 / ERAMER
cytosol / ER motifs

ERAP2+NPDP5

TPP2

PREP

+35 more in CSV

MHC-I candidate ligands

C01:02 0.04%

B27:05 0.25%

C07:01 0.058%

B08:01 0.096%

K | T | V | R | I | Y | Q | G | R | V | V | P | F | T | K | A | L | M | I | K | F | E | E | I

MHC-II candidate ligands

DQ04:01/04:02 1.1%

DQ05:01/02:01 4.4%

DR08:01 3.5%

ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward

NetCleave II native 0-1

0.5 display

SERUM / EXTRACELLULAR - native scores and recognition motifs

DPP4 native (intact SLP)

0.49

serum / extracellular motifs

MME

MME

MME

MME

MME

MME

ACE

MME

SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

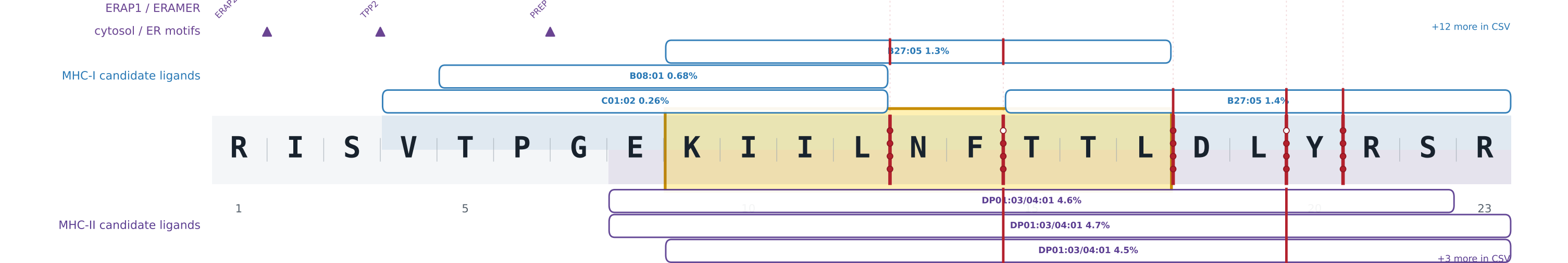
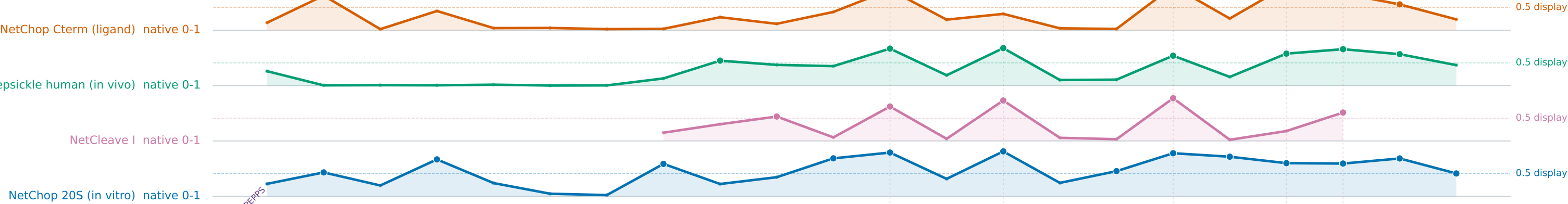
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

BMP1 p.Ser366Thr | JLF V3 | 23 aa

disclosed minimal epitope KIILNFTTL at 9-17

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

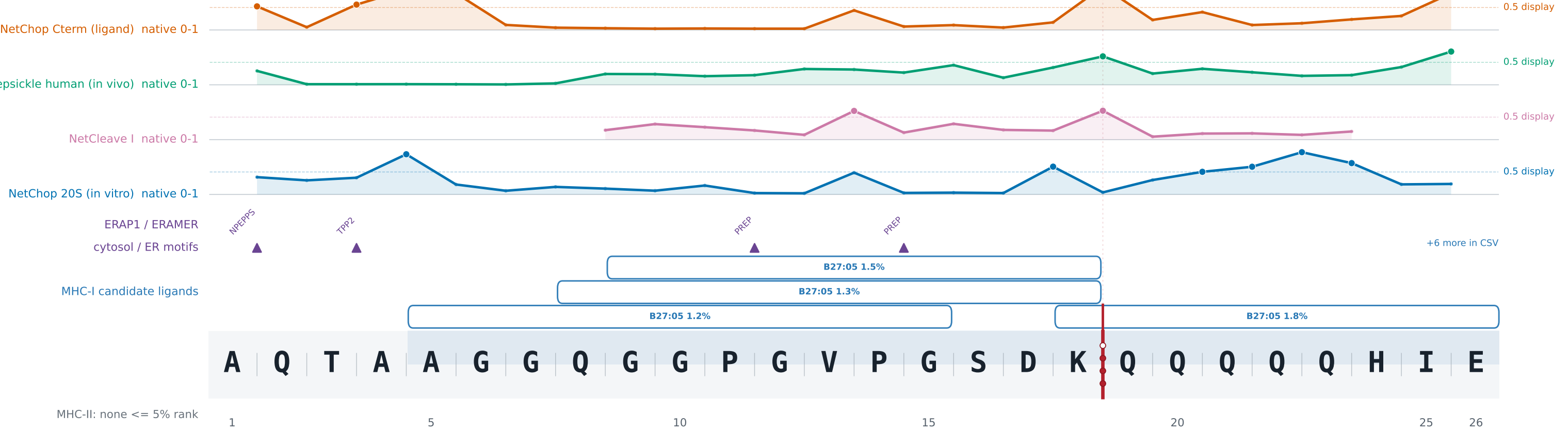
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

NR2F2 p.Gly65Val | JLF V2 | 26 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

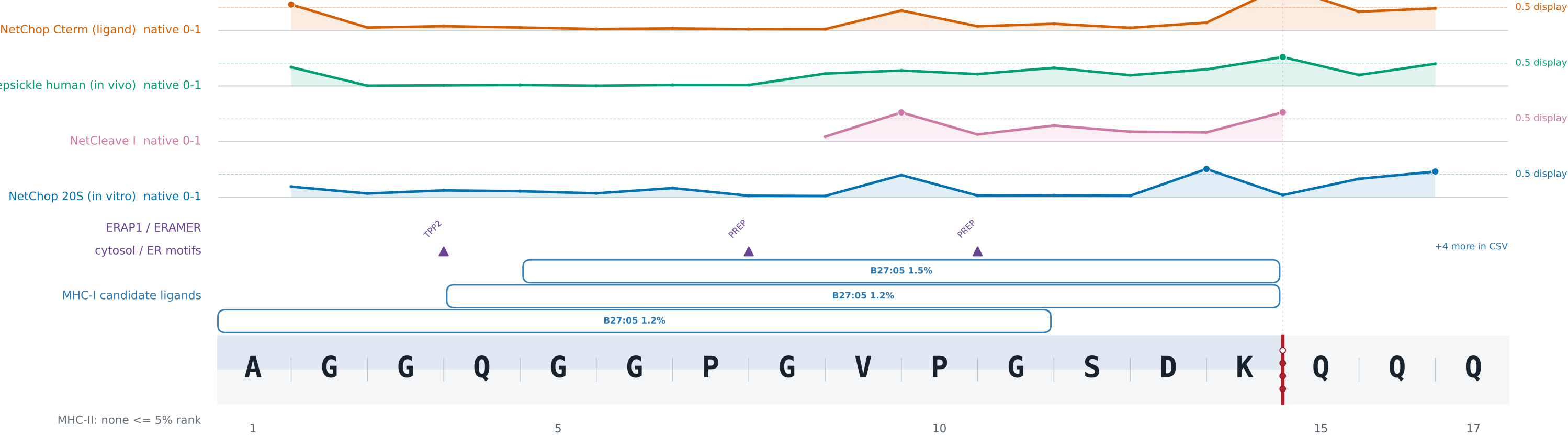
MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

NR2F2 p.Gly65Val | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks

higher native score upward



ENDOLYSOSOMAL / CLASS-II PROCESSING - C-terminal score track

higher native score downward



SERUM / EXTRACELLULAR - native scores and recognition motifs



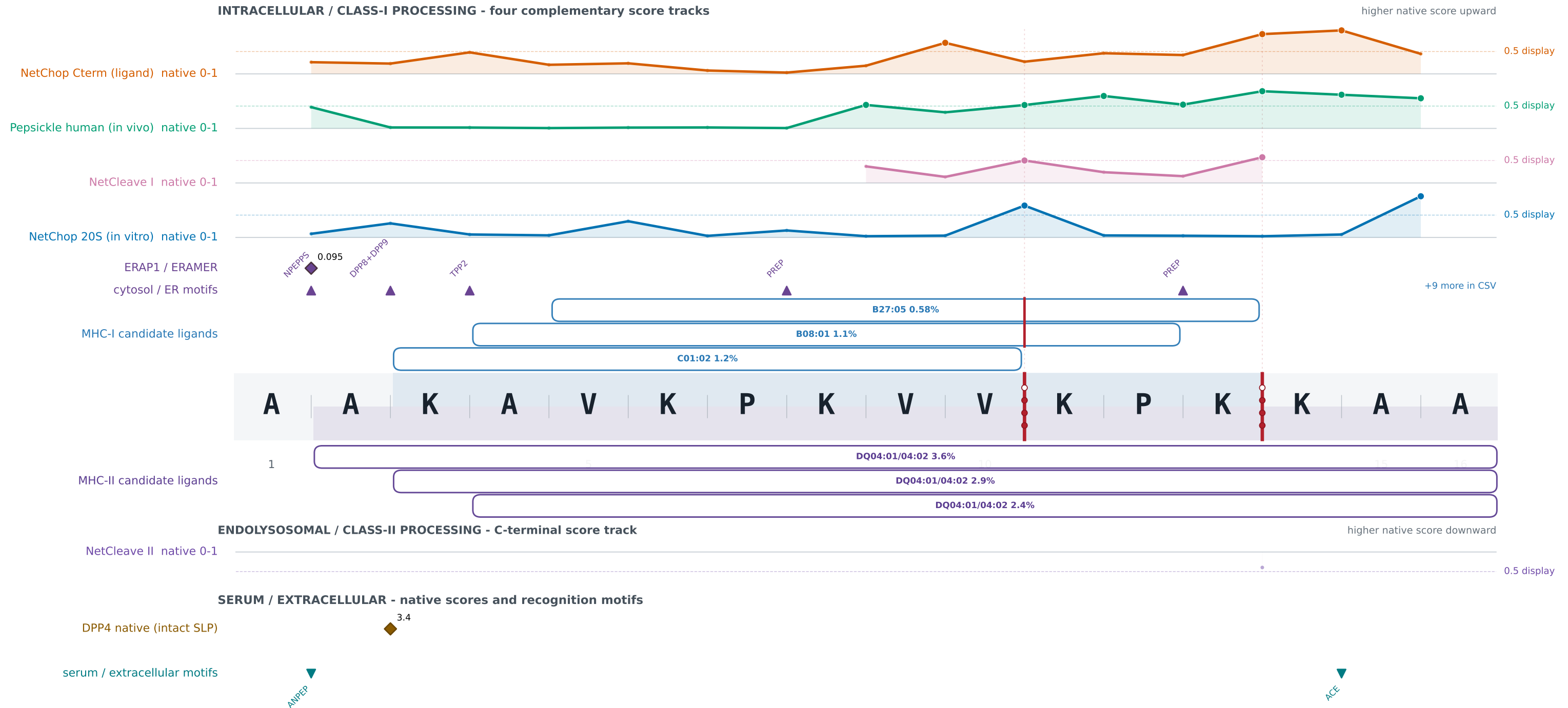
SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; >=3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.

H1-2 p.Ala197_Lys201del | CeGaT | 16 aa

minimal epitope/window not disclosed for this SLP



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED BOND MARKS All four intracellular tracks assess the bond; ≥ 3 reach 0.5. Four beads show support (filled = hit, open = miss), not probability.

MHC WINDOWS <=2%/<=5% rank; display favors intended overlap, then allele diversity, then rank. Internal red ticks are pre-binding cut evidence; all predictions remain in CSV.